

# Sixing Chen

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## EDUCATION

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### New York University

*Ph.D. in Cognition & Perception*

New York, NY

Sep. 2023 – Present

### Peking University

*B.S. in Cognitive Science*

Beijing, China

Sep. 2018 – May. 2023

## EXPERIENCE

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### AI Safety Research Fellow, Cambridge Boston Alignment Initiative

Jun. 2026 – Present

Mentor: Natalie Shapira

- Identified circuit-level causal mechanisms underlying sycophancy in LLMs.
- Developed an inference-time causal intervention method that eliminates sycophancy in LLMs.

### Doctoral Researcher, New York University

Sep. 2023 – Present

Advisor: Marcelo Mattar

- Investigated whether and how chain-of-thought reasoning implements planning in LLMs during large-scale strategic gameplay.
- Built deep reinforcement learning agents to model and predict eye-tracking data and single-neuron recordings during reasoning and decision-making.
- Developed a mathematical theory explaining how efficient planning can be implemented by mental simulation.

### Undergraduate Research Assistant, Georgia Institute of Technology

Jan. 2022 – May 2023

Advisor: Dobromir Rahnev

- Designed behavioral experiments to study how cognitive factors shape perceptual decision-making.
- Analyzed large-scale behavioral datasets to characterize how humans report confidence in their decisions.

### Undergraduate Research Assistant, Peking University

Sep. 2020 – May 2023

Advisor: Hang Zhang

- Developed computational models of how humans represent temporal uncertainty and how those representations shape motor preparation.

## PUBLICATIONS & PREPRINTS

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**Chen, S.**, Ying, Z. J., Smith, L. R., Wertheimer, J., & Shapira, N. (2026). Tracing mechanisms of sycophantic agreement in language models. *arXiv*.

**Chen, S.**, Li, J. A., Cakir, S., Akcali, S., Lee, K., & Mattar, M. G. (2026). Extracting Search Trees from LLM Reasoning Traces Reveals Myopic Planning. *Advances in Neural Information Processing Systems (NeurIPS)*.

**Chen, S.\***, Callaway, F.\*, Kumar, S., Lupkin, S. M., Wallis, J. D., McGinty, V. B., Rich, E. L., & Mattar, M. G. (2026). Learning to select computations in recurrent neural circuits. *bioRxiv*.

**Chen, S.**, Bocheva, M., & Rahnev, D. (2026). Signatures proposed to index perceptual effects emerge in a purely cognitive task. *Psychonomic Bulletin & Review*, 33(3), 75.

Nicholas, J., **Chen, S.**, & Mattar, M. G. (2026). Flexible decisions arise from resource-rational memory sampling. *bioRxiv*.

**Chen, S.\***, Jensen, K. T.\*, & Mattar, M. G. (2025). Rational decisions in multi-step environments with few rollouts. *PsyArXiv*.

Gao, Y., **Chen, S.**, & Rahnev, D. (2025). Dynamics of sensory and decisional biases in perceptual decision making: Insights from the face distortion illusion. *Psychonomic Bulletin & Review*, 32(1), 317–325.

**Chen, S.**, Jensen, K. T., & Mattar, M. G. (2024). Some and done? Temporally extended decisions with very few rollouts. *Proceedings of the Annual Meeting of the Cognitive Science Society*, Vol. 46.

**Chen, S.**, & Rahnev, D. (2023). Confidence response times: Challenging postdecisional models of confidence. *Journal of Vision*, 23(7), 11–11.

## CONFERENCE PRESENTATIONS

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**Chen, S.**, Callaway, F., Kumar, S., Wallis, J. D., Rich, E. L., & Mattar, M. G. (Aug. 2026). Learning to select computations in recurrent circuits. *Cognitive Computational Neuroscience (CCN)*, New York, NY, United States.

**Chen, S.**, Callaway, F., Kumar, S., Wallis, J. D., Rich, E. L., & Mattar, M. G. (Apr. 2026). Learning to select computations in recurrent circuits. *Manhattan Area Learning Meeting (MALM)*, New York, NY, United States. [Oral Presentation]

**Chen, S.**, Callaway, F., Kumar, S., Wallis, J. D., Rich, E. L., & Mattar, M. G. (Mar. 2026). Learning to select computations in recurrent circuits. *Computational and Systems Neuroscience (COSYNE)*, Lisbon, Portugal.

**Chen, S.**, Callaway, F., Kumar, S., & Mattar, M. G. (Jun. 2025). Meta-learning of human-like planning strategies. *Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM)*, Dublin, Ireland. [Oral Presentation]

**Chen, S.**, Jensen, K. T., & Mattar, M. G. (Jul. 2024). Some and done? Temporally extended decisions with very few rollouts. *Annual Meeting of the Cognitive Science Society (CogSci)*, Rotterdam, Netherlands.

**Chen, S.**, Jensen, K. T., & Mattar, M. G. (Mar. 2024). Some and done? Temporally extended decisions with very few rollouts. *Computational and Systems Neuroscience (COSYNE)*, Lisbon, Portugal.

**Chen, S.**, Wang, M., & Zhang, H. (Nov. 2022). The alternation of human temporal beliefs between different possibilities. *Society for Neuroscience (SfN)*, San Diego, CA, United States.

## TEACHING EXPERIENCE

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New York University

Teaching Assistant, Computational Programming in Psychological Sciences Feb. 2026 – May. 2026

Teaching Assistant, Lab for Cognition and Perception Feb. 2025 – May. 2025

Peking University

Teaching Assistant, Computational Modeling for Psychology and Neuroscience Feb. 2022 – Jun. 2022

## AWARDS & SCHOLARSHIPS

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AI Safety Research Fellowship, Cambridge Boston Alignment Initiative 2026

Engberg Fellowship, New York University 2023

Fenjiu Group Public Scholarship, Peking University 2021

## TECHNICAL SKILLS

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**Programming:** Python, JavaScript, R, MATLAB

**ML / DL:** PyTorch, TensorFlow, scikit-learn

**LLM:** vLLM, HuggingFace Transformers, NNSight

**Tools:** Git, Linux, SLURM, Jupyter